Topics and how they change over time

```
tx <- list(
  seed = 2026L,
  min_docfreq = 5L,  # drop words in fewer paragraphs than this
  run_searchK = TRUE,  # TRUE once to choose K (slow)
  K_grid = seq(2, 40, 2),
  K = 12L,  # set from the searchK plots
  spline_df = 5L,  # flexibility of the year trend
  run_sensitivity = TRUE,  # TRUE once to test the merge settings
  sens_factors = c(0.75, 1.25),  # multiples of min_tokens and max_tokens
  n_words = 20L,  # words per topic shown to the codebook model
  n_examples = 5L,  # example paragraphs per topic shown to the codebook model
  n_draws = 25L,  # draws of topic shares behind each trend interval
  workers = 20L,
  use_openrouter = TRUE,  # FALSE at work
  steps = tibble::tribble(
    ~step, ~base_url_env, ~api_key_env, ~model,
    "codebook", "COMPASS_LARGE_URL", "COMPASS_LARGE_KEY", "LARGE-MODEL"))

suppressPackageStartupMessages({
  library(stm); library(splines); library(viridis)
  library(tidyverse)  # tidyverse last so its verbs win
})
theme_set(theme_minimal(base_size = 12))
set.seed(tx$seed)
source("../topics_source.R")  # shared functions, one folder up
if (isTRUE(tx$use_openrouter)) source("../llm_openrouter.R")
```

Input

```
para_in <- p_out("paragraphs.csv")
paragraphs <- readr::read_csv(para_in, show_col_types = FALSE,
                              col_types = readr::cols(id = "c", para_uid = "c"))
writeLines(glue::glue("{nrow(paragraphs)} paragraph units from \\
  {dplyr::n_distinct(paragraphs$id)} comments."))
```

951 paragraph units from 60 comments.

Document-term matrix

Words are lowercased and kept only if purely alphabetic. Stopwords and month and weekday names are removed; temporal words would let topics encode the year directly. Words in fewer than 5 paragraphs are dropped.

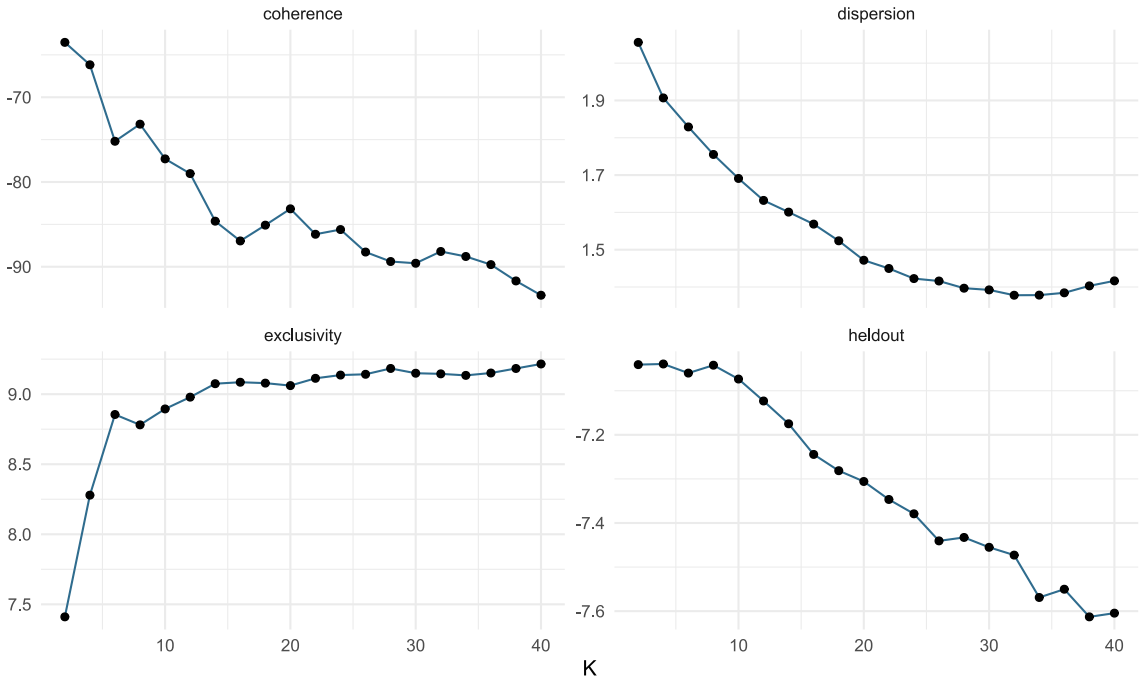
```
s <- stm_setup(paragraphs, tx$min_docfreq, tx$spline_df)
glue::glue("{length(s$documents)} paragraphs, {length(s$vocab)} words")
```

```
951 paragraphs, 2517 words
```

Choosing K

Take the smallest K where held-out likelihood has flattened and residual dispersion has stopped falling, then break ties on coherence and exclusivity. If the best values sit at the edge of the grid, widen it.

```
if (isTRUE(tx$run_searchK)) {
  cached(p_out("searchK.parquet"), para_in, search_k(s, tx$K_grid, tx$seed,
tx$workers)) |>
    plot_search_k()
} else glue::glue("searchK skipped; K = {tx$K}")
```



Structural topic model

Year enters topic prevalence as a spline, so each topic's share can rise and fall. Spectral initialization makes the fit reproducible. The fit is saved as `stm_fit.rds` and reloaded later; delete it to refit.

```
```

```
stm_fit <- cached(p_out("stm_fit.rds"), para_in, fit_stm(s, tx$K))
K <- ncol(stm_fit$theta)
if (K != tx$K) stop("stm_fit.rds has K = ", K, " but tx$K = ", tx$K,
                    ". Delete stm_fit.rds, then re-render.")
```

Assigning paragraphs to topics

Each paragraph is assigned its most likely topic. **Entropy R^2** (0 to 1) says how sharply paragraphs sort into topics overall; **AvePP** is the average topic share among a topic's own paragraphs, with about 0.7 the usual floor. Both fall as units get longer, so use them to compare topics within a fit, not to choose the unit size.

```
tables <- topic_tables(stm_fit, s$meta,
                    c("para_uid", "id", "para_id", "Year", "n_tokens", "text"),
tx$n_words)
theta_in <- p_out("paragraph_theta.csv")
readr::write_csv(tables$paragraph_theta, theta_in)
readr::write_csv(tables$topic_summary, p_out("topic_summary.csv"))

writeLines(glue::glue("Entropy R2 = {round(tables$entropy_r2, 3)}. Topics with
AvePP \\
  under 0.7: {sum(tables$topic_summary$avepp < 0.7, na.rm = TRUE)} of {K}."))
```

Entropy R2 = 0.591. Topics with AvePP under 0.7: 11 of 12.

```
tables$topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::select(topic, prevalence, n_paras, avepp) |>
  tbl_paged(caption = "Topics by prevalence.")
```

Topics by prevalence.

topic	prevalence	n_paras	avepp
7	0.136	142	0.655
10	0.100	104	0.647
5	0.092	80	0.647
6	0.089	79	0.711
4	0.087	91	0.633
2	0.079	81	0.629
8	0.075	76	0.592
9	0.074	60	0.694
3	0.072	69	0.609
1	0.069	58	0.630
12	0.068	62	0.642
11	0.058	49	0.652

Do the merge settings drive the topics?

The paragraph units are rebuilt at 0.75 and 1.25 times `min_tokens` and `max_tokens` from the saved embeddings (no API calls), and the model is refit at the same K with the same year spline. Each baseline topic is matched to its closest topic in each refit by the cosine of their word distributions. A topic with a low cosine exists only at one way of cutting the text and deserves caution.

```
if (isTRUE(tx$run_sensitivity)) {
  sens_in <- p_out(c("paragraphs.csv", "blocks.parquet", "embeddings.parquet",
                     "merge_settings.parquet"))
  blocks <- arrow::read_parquet(sens_in[2])
  E <- emb_matrix(blocks, arrow::read_parquet(sens_in[3]))
  ms <- arrow::read_parquet(sens_in[4])
  refit <- function(f) {
    p <- link_corpus(blocks, E, ms$min_tokens * f, ms$max_tokens * f, ms$min_sim)
|>
      dplyr::filter(keep)
    alt <- stm_setup(p, tx$min_docfreq, tx$spline_df)
    # same spline knots as the baseline, so only the cut of the text differs
    alt$meta[colnames(s$bs)] <- stats::predict(s$bs, newx = alt$meta$Year)
    match_topics(beta_matrix(stm_fit), beta_matrix(fit_stm(alt, K))) |>
      dplyr::mutate(factor = f, paragraphs = length(alt$documents))
  }
  sens_tbl <- cached(p_out("sensitivity.parquet"), sens_in,
                     purrr::list_rbind(purrr::map(tx$sens_factors, refit)))
  if (max(sens_tbl$topic) != K)
    stop("sensitivity.parquet was built at another K. Delete it and re-render.")
```

```
  sens_tbl |>
    dplyr::group_by(factor) |>
    dplyr::summarise(paragraphs = dplyr::first(paragraphs),
                     median_cosine = stats::median(cosine), lowest_cosine =
min(cosine)) |>
    tbl_paged(caption = "Matched topic cosine against the baseline fit.")
  sens_tbl |>
      tidyr::pivot_wider(id_cols = topic, names_from = factor, values_from =
cosine,
                         names_prefix = "x") |>
    tbl_paged(caption = "Matched cosine per topic.")
} else writeLines("Sensitivity test skipped (tx$run_sensitivity = FALSE).")
```

Matched topic cosine against the baseline fit.

factor	paragraphs	median_cosine	lowest_cosine
0.75	1129	0.797	0.415
1.25	814	0.765	0.549

Matched cosine per topic.

topic	x0.75	x1.25
1	0.582	0.819
2	0.799	0.549
3	0.828	0.748
4	0.774	0.930
5	0.586	0.752
6	0.844	0.950
7	0.835	0.669
8	0.794	0.788
9	0.892	0.814
10	0.934	0.702
11	0.790	0.704
12	0.415	0.779

Topic codebook

One call per topic, in parallel, with the larger model. Each call sees only that topic's 20 most distinctive (FREX) and most probable words and its 5 most typical paragraphs. It drafts a label, a

one-sentence description, and a proposition for the stance app. If `topic_codebook.csv` exists it is read, not redrafted, so your edits survive. Delete it to redraft.

```
codebook <- draft_codebook(p_out("topic_codebook.csv"), theta_in, tables,
"public comments",
                           tx$steps, tx$use_openrouter, tx$n_examples)
dup <- codebook |> dplyr::add_count(label) |> dplyr::filter(n > 1, !is.na(label))
if (nrow(dup)) writeLines(glue::glue("**Duplicate labels to fix:** \\
  {str_c('topic ', dup$topic, ' (', dup$label, ')', collapse = ', ')}."))
codebook |>
    dplyr::left_join(dplyr::select(tables$topic_summary, topic, frex), by =
"topic") |>
  dplyr::mutate(title = str_c("Topic ", topic, ": ", label)) |>
  tbl_records("title", c("description", "proposition", "frex"),
              caption = "Topic codebook. Edit the CSV; nothing redrafts it.")
```

Topic codebook. Edit the CSV; nothing redrafts it.

Topic 1: Civil War Context - description: The paragraphs discuss the context and causes of a civil war, the state of the Union, and the role of divine providence in resolving national differences. - proposition: The nation's security depends on its acknowledged dependence upon God. - frex: let, differences, religious, around, divide, earth, capacity, faculties, british, physical, surely, war, fathers, rest, prevail, unite, still, five, present, man

Topic 2: National Defense Commerce - description: The paragraphs discuss national defense, economy, and commerce, including the role of the government in protecting and promoting the country's industry and trade. - proposition: The government should protect national industry through tariffs. - frex: revenue, currency, army, debt, trade, navy, credit, south, naval, coast, commerce, necessary, industry, tariff, manufactures, invasion, agriculture, taxes, products, pay

Topic 3: Executive Office Assumption - description: The paragraphs discuss the assumption of executive office, the duties associated with it, and the speaker's gratitude toward the citizens. - proposition: The task of the first executive office is above the talents of the person undertaking it. - frex: discharge, gratitude, duties, station, countrymen, solemn, almighty, presence, blessed, sustain, enter, trust, generous, occasion, hopes, beloved, high, providence, devotion, arduous

Topic 4: Federal Power Limits - description: The paragraphs discuss the distribution and limits of power among different levels of government and institutions within the United States. - proposition: The Constitution clearly defines the limits of power for the general government and the states. - frex: union, territory, confederacy, constitution, granted, territories, general, grant, sovereignty, settled, powers, states, exercised, domestic, texas, authority, constitutional, blessings, separate, institutions

Topic 5: Government and Economy - description: The paragraphs discuss the role of government and economy in ensuring the well-being of citizens, particularly the vulnerable, and addressing issues like poverty and inequality. - proposition: A great nation must prioritize the needs

of its most vulnerable citizens. - frex: women, work, families, children, health, wealth, lives, something, know, economy, society, succeed, built, forces, schools, workers, land, child, rich, cost

Topic 6: International Cooperation and Freedom - description: The paragraphs discuss international cooperation, the defense of freedom, and the role of the United Nations in maintaining peace. - proposition: The United Nations is the last best hope for peace in an age of war. - frex: peoples, seek, weapons, help, program, peace, nations, helped, world, threat, freedom, worlds, nuclear, strive, strong, enemies, free, price, areas, strength

Topic 7: American National Renewal - description: The paragraphs discuss the renewal of America through service, hope, and the continuation of the American story. - proposition: A new generation of Americans should engage in a season of service. - frex: americas, america, promise, heart, americans, journey, story, generation, let, century, dignity, together, generations, challenges, forward, voices, dreams, us, challenge, resolve

Topic 8: American National Ideals - description: American national ideals and international relations. - proposition: The United States desires territorial expansion for economic domination of other peoples. - frex: self, civilization, thought, ideals, understanding, problems, republic, economic, vision, foundations, wish, fail, lies, aided, realization, profit, mankind, life, courage, counsel

Topic 9: Foreign Policy Principles - description: The paragraphs discuss foreign policy and international relations. - proposition: The United States should cultivate friendly relations with all nations. - frex: relations, foreign, policy, press, rights, suffrage, friendly, respect, settlement, nations, friendship, universal, advantage, obligations, cultivate, ballot, arbitration, intercourse, controversies, ignorance

Topic 10: Government Reform Needed - description: The paragraphs discuss government reform and the need for changes in various aspects of public administration and law enforcement. - proposition: Civil service reform should be thorough and radical. - frex: business, enforce, laws, amendment, enforcement, legislation, partisan, expenditures, labor, interstate, due, money, public, session, official, combinations, taxation, reform, popular, election

Topic 11: Separation of Powers - description: The paragraphs discuss the separation of powers and checks and balances among the branches of government. - proposition: The Constitution intended for the Executive branch to play a significant role in legislation. - frex: measures, legislature, executive, distinguished, former, conduct, heretofore, causes, enlightened, certainly, contest, occurred, department, instance, early, circumstances, formed, least, although, opinion

Topic 12: Presidential Inauguration Speeches - description: The paragraphs discuss inauguration ceremonies of US presidents. - proposition: Inauguration ceremonies celebrate the unity of the country. - frex: oath, president, thank, vice, day, mr, ago, going, taken, ceremony, dont, years, four, today, washington, inauguration, celebrate, chief, first, accept

Prevalence over time

For each topic, its share of each paragraph is regressed on the year spline with a fractional logit (Papke and Wooldridge 1996), so every estimated share stays between 0 and 1. The model is refit

on 25 draws of the topic shares and the results combined, so the 95 percent intervals include the topic model's own uncertainty and the clustering of paragraphs within a comment.

```
trends <- topic_trends(stm_fit, s, tx$n_draws, tx$seed, s$meta$id)
readr::write_csv(trends, p_out("topic_trends.csv"))
```

```
trends |>
  dplyr::left_join(dplyr::select(codebook, topic, label), by = "topic") |>
  dplyr::left_join(dplyr::select(tables$topic_summary, topic, prevalence), by =
"topic") |>
  plot_trends()
```

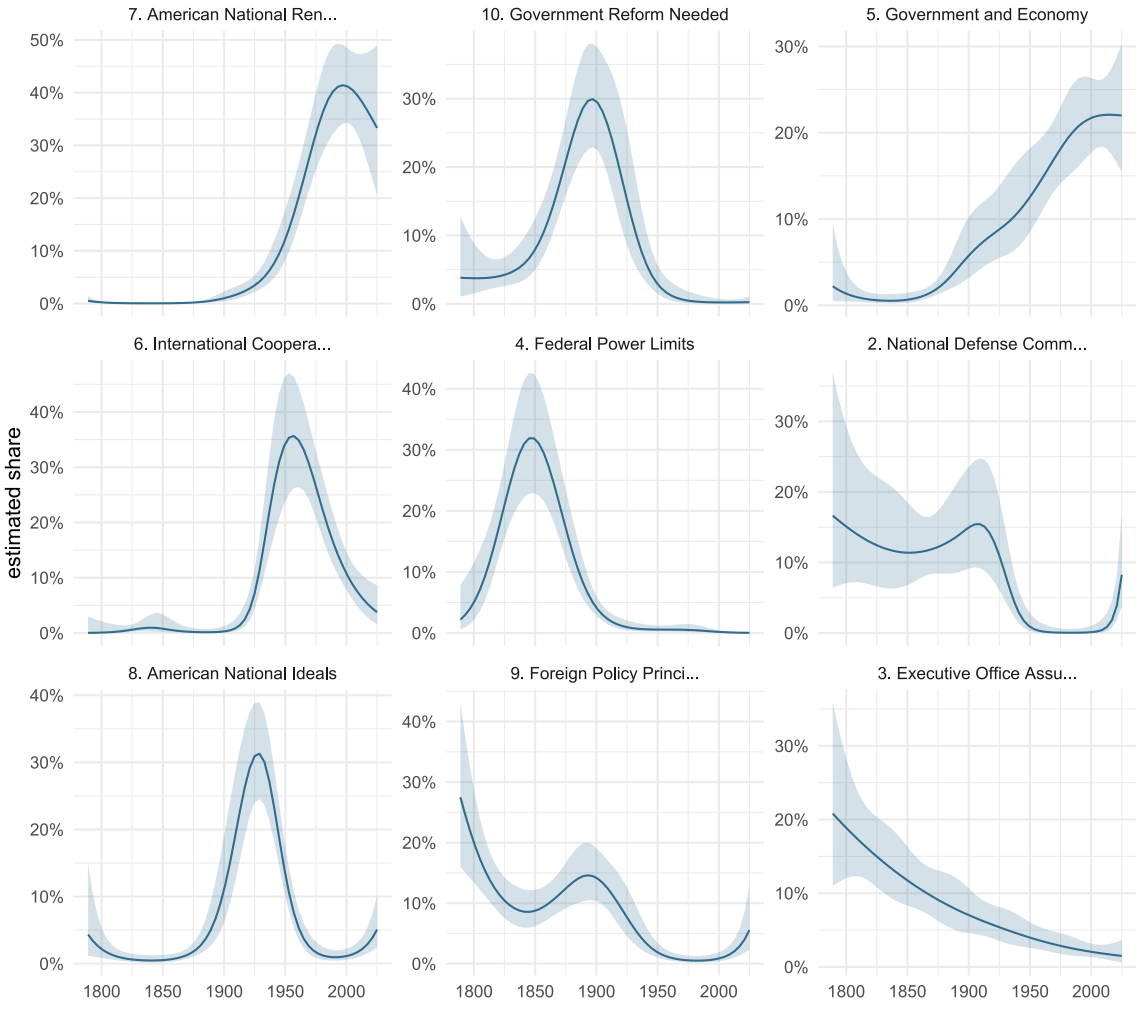


Figure 1: Expected topic share by year with 95 percent intervals, nine topics per page, most prevalent first.

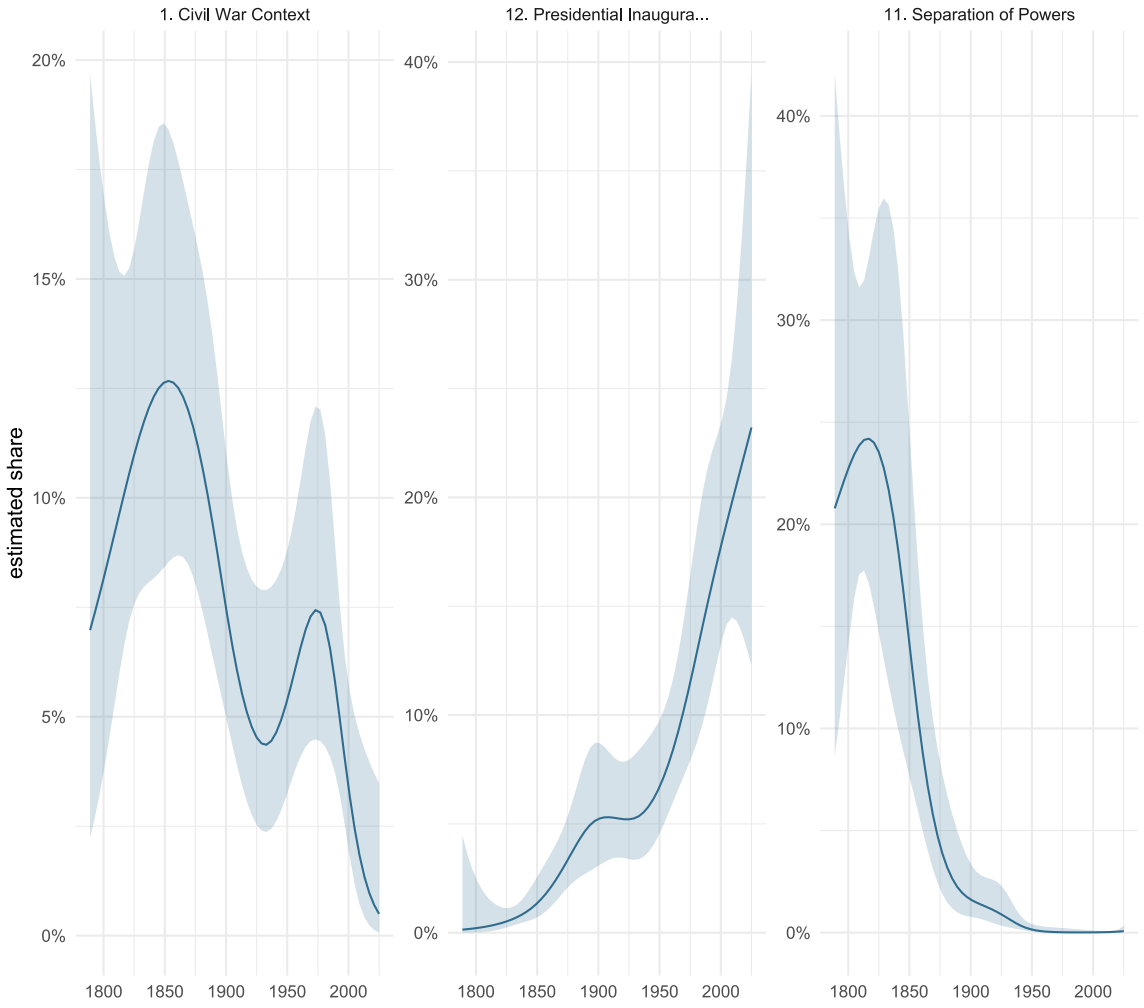


Figure 2: Expected topic share by year with 95 percent intervals, nine topics per page, most prevalent first.

```
writeLines(glue::glue("Wrote paragraph_theta.csv, topic_summary.csv,
topic_codebook.csv, \\
  topic_trends.csv and stm_fit.rds to topics/comments/outputs/."))
```

Wrote paragraph_theta.csv, topic_summary.csv, topic_codebook.csv, topic_trends.csv and stm_-fit.rds to topics/comments/outputs/.